

AI Invoice Processing Agent

User Guide

Agentic AI-based invoice automation using n8n, Google Gemini and Google Workspace

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1. Introduction

Project Overview

The **AI Invoice Processing Agent** is an Agentic AI-based business process automation solution designed to automate invoice processing.

The system automatically receives invoice PDFs through **Gmail**, extracts structured invoice information using **Google Gemini**, validates invoice data, detects duplicate invoices, matches invoices with **Purchase Orders (POs)**, applies predefined business rules, records processing results, and sends automated notifications.

The solution also uses a **Human-in-the-Loop** approach to route invoices requiring manual review instead of automatically processing exceptions.

Objective

The primary objective is to automate the invoice processing lifecycle while maintaining **validation, traceability, business-rule control, and human oversight** for exceptions.

2. System Architecture

Components Used

Component	Purpose
n8n Cloud	Workflow automation and orchestration
Gmail	Invoice email trigger and processing notifications
Google Drive	Storage of original invoice PDFs
Google Gemini	AI-based invoice information extraction
Structured Output Parser	Converts AI output into structured JSON
JavaScript / n8n Code	Validation, duplicate detection and decision logic
Google Sheets	PO Master and Invoice Processing Log
Human-in-the-Loop	Exception handling and manual review

Workflow Overview

1. Invoice PDF is received through Gmail.
2. The invoice email and attachment are retrieved.
3. The original invoice PDF is stored in Google Drive.
4. Invoice text is extracted from the PDF.
5. Google Gemini extracts structured invoice information.
6. Extracted invoice data is validated.
7. The system checks for duplicate invoices.
8. The invoice is matched with the corresponding Purchase Order.

9. Deterministic business rules determine the final decision.
10. The processing result is stored in Google Sheets.
11. A processing notification or human-review alert is sent through Gmail.

3. Data Sources

PO Master

The **PO_Master** sheet contains Purchase Order information used for invoice matching.

Main fields: PO Number, Vendor Name, PO Amount, Currency, PO Status.

Invoice Log

The **Invoice_Log** sheet stores the results of processed invoices.

Main fields: Invoice Number, Vendor Name, Invoice Date, Due Date, PO Number, Subtotal, Tax, Total Amount, Currency, Validation Status, Validation Issues, PO Match Status, Decision, Reason, Processed Date.

4. Accessing the System

The system operates through the configured **n8n Cloud** workflow. Users do **not** need to enter invoice information manually.

Required Input

The user only needs to send an invoice through the configured Gmail account.

Invoice Requirements

- Invoice should be provided as a **PDF attachment**.
- Invoice should contain the relevant invoice information.
- A Purchase Order number should be available when PO matching is required.

Basic Usage

1. Open the configured Gmail account.
2. Create an email containing the invoice PDF as an attachment.
3. Send the email to the configured invoice-processing Gmail address.
4. The Gmail trigger automatically starts the n8n workflow.
5. The system processes the invoice automatically.

5. Invoice Processing Workflow

After receiving an invoice, the system performs the following operations.

Step	Stage	Description
1	Invoice Email Trigger	Gmail trigger detects the incoming invoice email.
2	Retrieve Invoice Email	Retrieves the email and downloads the invoice attachment.
3	Store Original Invoice	Uploads the original PDF to Google Drive for storage and traceability.
4	Extract Invoice Text	Converts PDF content into processable text using n8n file extraction.
5	AI Invoice Data Extraction	Google Gemini analyzes the text and produces structured invoice information.

6. AI Invoice Data Extraction

Google Gemini converts unstructured invoice information into **structured JSON** data.

Information Extracted

Invoice Number, Vendor Name, Invoice Date, Due Date, PO Number, Subtotal, Tax, Total Amount, Currency.

Example

Invoice Number : INV-1008
Vendor : ABC Supplies Pvt Ltd
Invoice Date : 2026-09-28
PO Number : PO-501
Subtotal : ₹40,000
Tax : ₹5,000
Total Amount : ₹50,000
Currency : INR

The **Structured Output Parser** ensures the AI response follows the required JSON structure.

Important design principle: The LLM extracts information from the invoice. It does *not* make the final approval decision.

7. Invoice Data Validation

After AI extraction, **deterministic validation rules** are applied.

Validation Checks

The system checks that the following exist:

- Invoice number
- Vendor name
- Invoice date
- PO number
- Subtotal
- Tax
- Total amount

Amount Validation

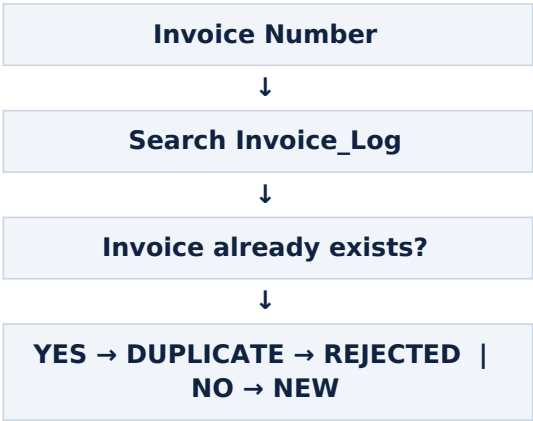
Subtotal + Tax = Total Amount

Subtotal = ₹40,000
Tax = ₹5,000
Expected Total = ₹45,000
Invoice Total = ₹50,000
Result = INVALID

An invoice that fails validation is routed for **Human Review**.

8. Duplicate Invoice Detection

The system checks whether the invoice has already been processed.



Duplicate Rule

If the invoice number already exists in the Invoice Log, then **Decision = REJECTED**. This prevents the same invoice from being processed multiple times.

9. PO & Vendor Validation

After duplicate checking, the system retrieves the corresponding Purchase Order from the **PO_Master** sheet.

Matching Criteria

PO Number, Vendor Name, Invoice Amount, Currency, PO Status.

Example

Field	Invoice	PO Master
PO Number	PO-501	PO-501
Vendor	ABC Supplies Pvt Ltd	ABC Supplies Pvt Ltd
Amount	₹45,000	₹45,000
Currency	INR	INR
Status	—	Approved

Result: PO Match Status = MATCHED. If vendor, amount, currency or other required PO conditions do not match, the invoice is routed to **Human Review**.

10. Invoice Decision Engine

The Decision Engine applies deterministic business rules to determine the final outcome.

Rule	Condition	Decision
1 — Duplicate invoice	Duplicate = YES	REJECTED
2 — Validation failure	Validation = INVALID	HUMAN REVIEW
3 — PO / vendor mismatch	PO Matching = MISMATCH	HUMAN REVIEW
4 — PO not approved	PO Status ≠ Approved	HUMAN REVIEW
5 — All checks pass	Validation = VALID, Duplicate = NEW, PO Match = MATCHED, PO Status = Approved	APPROVED

Possible Decisions

Decision	Meaning
APPROVED	All required checks passed
HUMAN REVIEW	An exception requires manual investigation
REJECTED	Duplicate invoice detected

11. Human-in-the-Loop

The system does **not** automatically approve invoices when an exception is detected. Instead, the invoice is routed to a human reviewer.

Human Review Conditions

- Invoice validation failure
- Amount mismatch
- Vendor mismatch
- Currency mismatch
- PO mismatch
- PO not approved
- Other business-rule exceptions

Human Review Process



This provides additional control over potentially incorrect or exceptional invoices.

12. Invoice Logging

After the decision is generated, the result is stored in the **Invoice_Log** Google Sheet.

Information Logged

Invoice Number, Vendor Name, Invoice Date, Due Date, PO Number, Subtotal, Tax, Total Amount, Currency, Validation Status, Validation Issues, PO Match Status, Decision, Reason, Processed Date.

The log provides a centralized record of processing activity and supports **traceability**.

13. Notification Process

The system automatically sends an email notification after processing.

Normal Processing Notification

Sent for invoices following the normal route. Contains: Invoice Number, Vendor, Invoice Date, PO Number, Invoice Amount, Validation Status, PO Match Status, Decision, Reason, Processing Date.

Human Review Alert

Sent when an invoice requires manual review. Contains: Invoice Number, Vendor, PO Number, Invoice Amount, Validation Status, Duplicate Status, PO Match Status, Decision, Reason, Validation Issues.

14. Google Drive Storage

The original invoice PDF is stored in Google Drive during processing.

Purpose

- Original invoice preservation
- Document traceability
- Easy access to processed invoice files
- Separation between source documents and processing results

The stored PDF can be referenced when a human reviewer investigates an exception.

15. Testing & Validation

The workflow was tested using multiple invoice scenarios.

Test Case	Scenario	Expected Result
TC-01	Valid invoice + matching PO	APPROVED
TC-02	Invoice amount $\neq$ PO amount	HUMAN REVIEW
TC-03	Duplicate invoice number	REJECTED
TC-04	Invoice arithmetic mismatch	HUMAN REVIEW

Testing Covered

Invoice extraction, structured AI output, data validation, duplicate detection, PO matching, vendor matching, amount matching, currency validation, decision logic, human-review routing, Gmail notifications, Google Sheets logging, Google Drive storage.

16. Key Controls & Design Principles

Control	Description
1. Structured AI Output	The parser constrains the Gemini response to the required JSON structure.
2. Deterministic Validation	Financial calculations use explicit business rules, not the LLM.
3. Duplicate Detection	Invoice numbers are checked against previously processed records.
4. PO Matching	Vendor, amount, currency and PO status are verified.
5. Human-in-the-Loop	Exceptions go to a human reviewer instead of being auto-approved.
6. Auditability	Results, decisions and reasons are stored in the Invoice Log.

Core design principle: *AI extracts information; deterministic business rules control the decision.*

17. Troubleshooting

Issue	Resolution
Invoice not processed	Verify the invoice was sent to the configured Gmail address, the email has an attachment, and the attachment is a PDF. Check the n8n execution history.
Attachment not extracted	Confirm the attachment exists and is a supported PDF. Check the Gmail retrieval step in n8n.
AI extraction error	Check the Gemini execution, confirm invoice text was extracted, verify Gemini credentials and model configuration, and review the AI Agent and Structured Output Parser output.
Sent to Human Review	Review validation issues, invoice arithmetic, PO amount, vendor name, currency and PO status. At least one business rule requires manual investigation.
Duplicate rejected	Search the Invoice_Log sheet by invoice number. If it exists, the invoice is a duplicate.
Notification not received	Verify the recipient address, check Spam/Junk, check the Gmail node execution in n8n, and verify Gmail credentials and permissions.

18. Limitations

- Processing depends on the availability of Gmail, Google Gemini, Google Drive and Google Sheets.
- AI extraction quality depends on the quality and readability of the invoice document.
- The workflow is designed around **PDF** invoices.
- Exceptions that fail predefined business rules require human review.
- PO matching depends on the availability and correctness of PO Master records.
- Google Sheets is used as the data repository instead of an enterprise ERP or database.

19. Future Enhancements

Integrations	Capabilities
ERP integration	OCR for scanned / image invoices
Vendor Master integration	Advanced approval thresholds
Accounts Payable integration	Multi-currency processing
Database-based invoice storage	Invoice analytics dashboard
Role-based access control	Fraud and anomaly detection
Automated payment workflow	Human approval interface

20. Project Deliverables

- Automated invoice email intake
- Original invoice storage
- AI-based data extraction with structured JSON output
- Invoice data, PO, vendor, amount and currency validation
- Duplicate invoice detection
- Deterministic decision engine
- Human-in-the-Loop exception handling
- Invoice processing log
- Automated Gmail notifications
- End-to-end workflow automation

21. Conclusion

The **AI Invoice Processing Agent** demonstrates how Agentic AI, workflow automation and deterministic business rules can be combined to automate a real business process, from invoice receipt to final processing decision.

- Receives invoices automatically
- Extracts information using Google Gemini
- Validates data and detects duplicates
- Matches invoices with Purchase Orders
- Applies deterministic business rules
- Logs results and stores original documents
- Sends automated notifications
- Routes exceptions to human reviewers

The solution provides a structured and controlled approach to invoice processing while maintaining validation, traceability and human oversight.